Address Mapping

Content

- ARP
- RARP
  - ICMP
  - IGMP

Introduction

- The delivery of a packet to a host or a router requires two levels of addressing: logical and physical.
- We need to be able to map a logical address to its corresponding physical address and vice versa.
- This can be done by using either static or dynamic mapping.

Physical addresses - change in the following ways

- A machine could change its NIC, resulting in a new physical address.
- In some LANs, such as LocalTalk, the physical address changes every time the computer is turned on.
- A mobile computer can move from one physical network to another, resulting in a change in its physical address.

Contd...

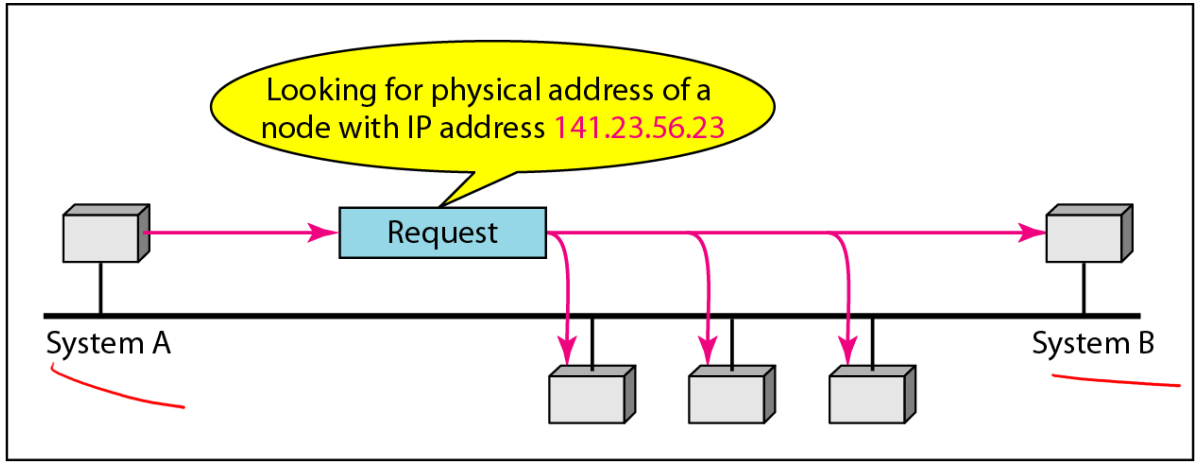
- To implement these changes, a static mapping table must be updated periodically.
- This overhead could affect network performance.
- In dynamic mapping each time a machine knows one of the two addresses (logical or physical), it can use a protocol to find the other one.

Mapping Logical to Physical Address: ARP

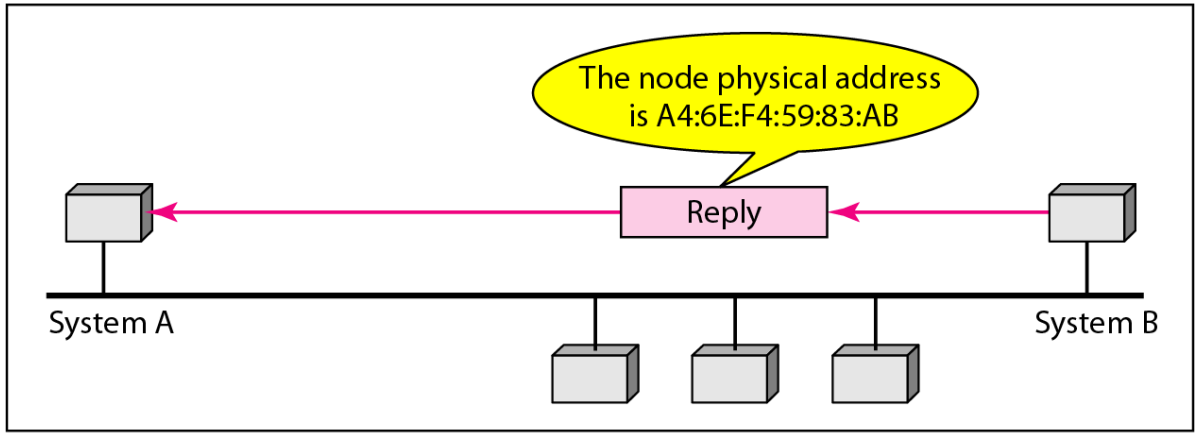
- ARP- Address Resolution Protocol

Mapping Logical to Physical Address

ARP (address resolution protocol)



a. ARP request is broadcast

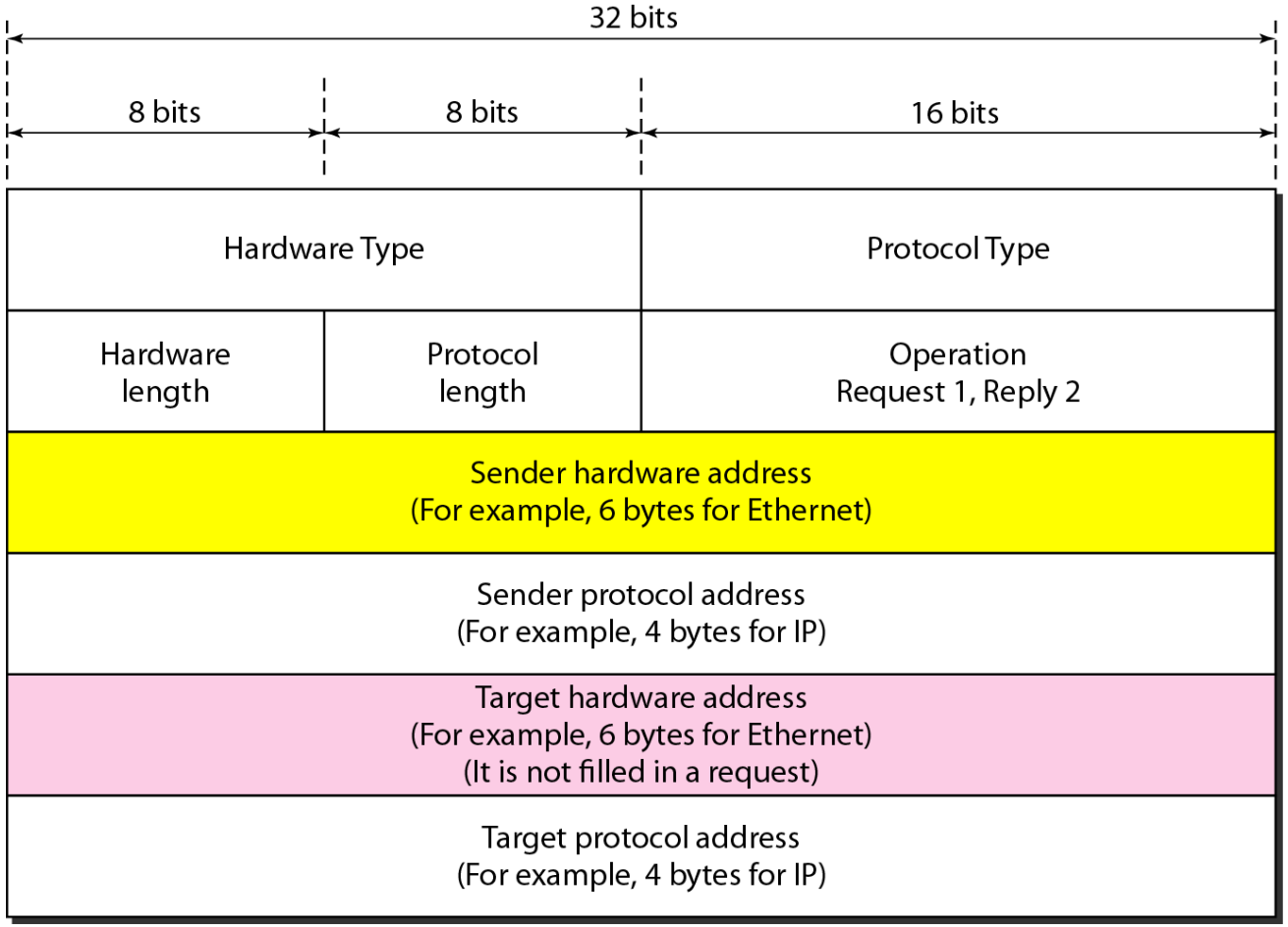


b. ARP reply is unicast

Note

ARP can be useful if the ARP reply is cached (kept in cache memory for a while).

ARP packet


32 bits
8 bits
8 bits
16 bits
Hardware Type
Protocol Type
Hardware length
Protocol length
Operation Request 1, Reply 2
Sender hardware address (For example, 6 bytes for Ethernet)
Sender protocol address (For example, 4 bytes for IP)
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled in a request)
Target protocol address (For example, 4 bytes for IP)

Contd...

- ***Hardware type:*** This is a 16-bit field defining the type of the network on which ARP is running. Each LAN has been assigned an integer based on its type. For example, Ethernet is given type 1. ARP can be used on any physical network.
- ***Protocol type:*** This is a 16-bit field defining the protocol. For example, the value of this field for the IPv4 protocol is 080016, ARP can be used with any higher-level protocol.
- ***Hardware length:*** This is an 8-bit field defining the length of the physical address in bytes. For example, for Ethernet the value is 6.

Contd…

- ***Protocol length:*** This is an 8-bit field defining the length of the logical address in bytes. For example, for the IPv4 protocol the value is 4.
- ***Operation:*** This is a 16-bit field defining the type of packet. Two packet types are defined: ARP request (1) andARP reply (2).
- ***Sender hardware address:*** This is a variable-length field defining the physical address of the sender. For example, for Ethernet this field is 6 bytes long.
- ***Sender protocol address:*** This is a variable-length field defining the logical (for example, IP) address of the sender. For the IP protocol, this field is 4 bytes long.

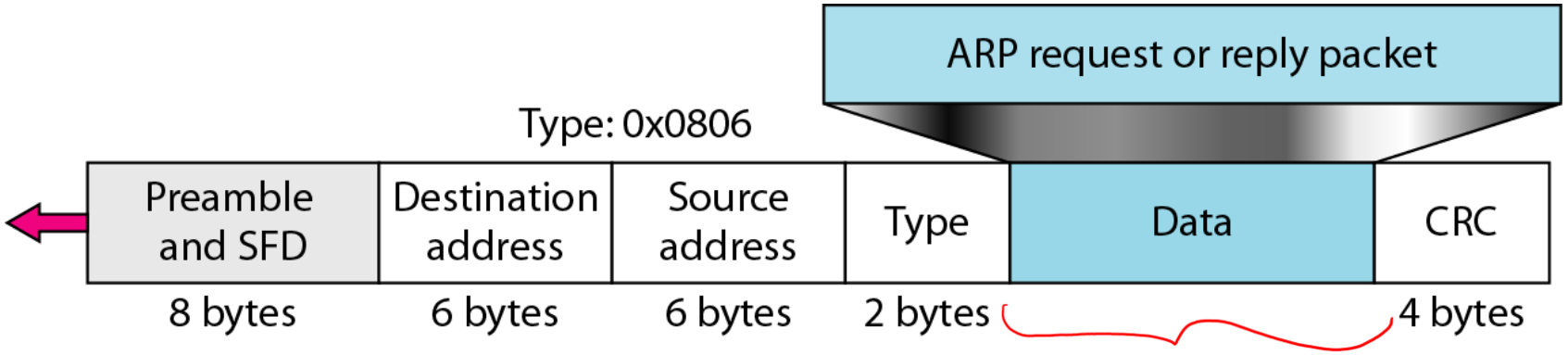
Contd...

- ***Target hardware address:*** This is a variable-length field defining the physical address of the target. For example, for Ethernet this field is 6 bytes long. For an ARP request message, this field is all Os because the sender does not know the physical address of the target.
- ***Target protocol address:*** This is a variable-length field defining the logical (for example, IP) address of the target. For the IPv4 protocol, this field is 4 bytes long.

Encapsulation

- An ARP packet is encapsulated directly into a data link frame.
- For example, in Figure an ARP packet is encapsulated in an Ethernet frame.
- The type field indicates that the data carried by the frame are an ARP packet.

Encapsulation of ARP packet



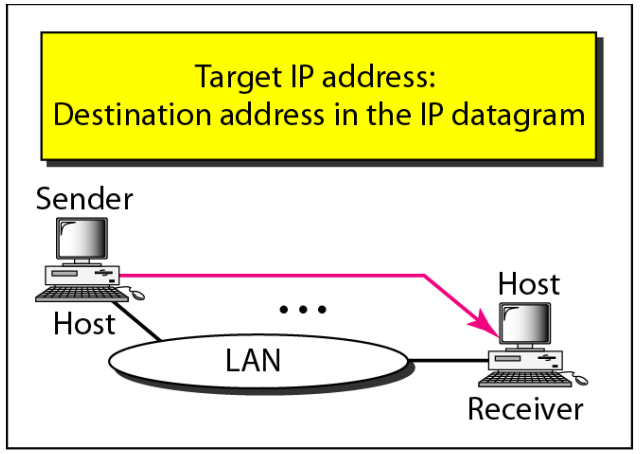
Steps involved in an ARP process

- The sender knows the IP address of the target. We will see how the sender obtains this shortly.
- IP asks ARP to create an ARP request message, filling in the sender physical address, the sender IP address, and the target IP address. The target physical address field is filled with Os.
- The message is passed to the data link layer where it is encapsulated in a frame by using the physical address of the sender as the source address and the physical broadcast address as the destination address.
- Every host or router receives the frame. Because the frame contains a broadcast destination address, all stations remove the message and pass it to ARP.

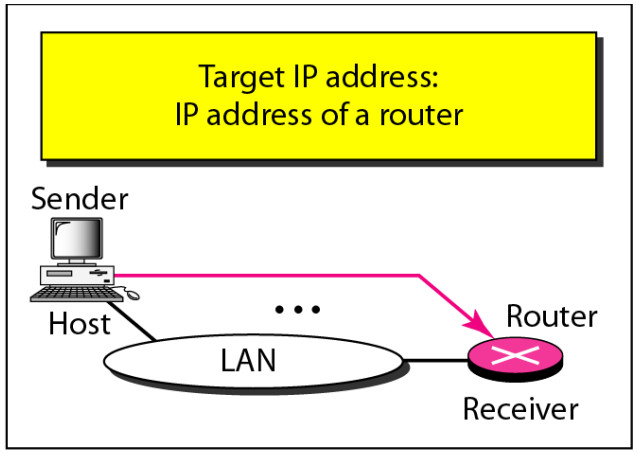
Contd...

- All machines except the one targeted drop the packet. The target machine recognizes its IP address.
- The target machine replies with an ARP reply message that contains its physical address. The message is unicast.
- The sender receives the reply message. It now knows the physical address of the target machine.
- The IP datagram, which carries data for the target machine, is now encapsulated in a frame and is unicast to the destination.

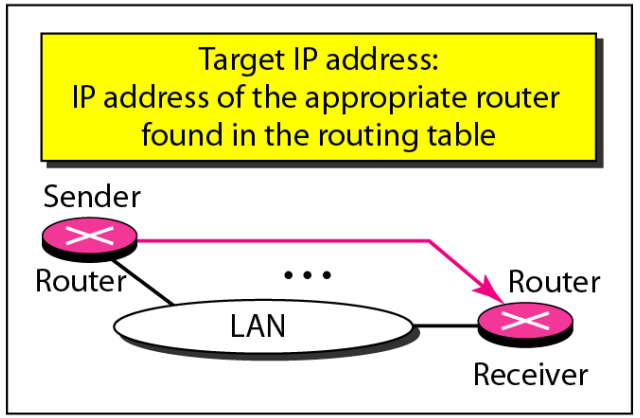
Four cases using ARP



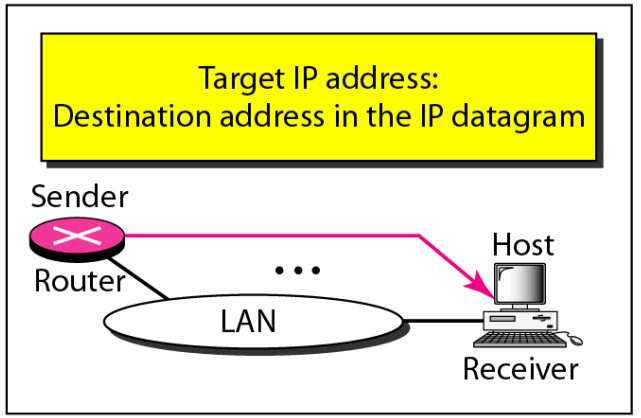
Case 1. A host has a packet to send to another host on the same network.



Case 2. A host wants to send a packet to another host on another network. It must first be delivered to a router.



Case 3. A router receives a packet to be sent to a host on another network. It must first be delivered to the appropriate router.



Case 4. A router receives a packet to be sent to a host on the same network.

Case 1

- The sender is a host and wants to send a packet to another host on the same network.
- In this case, the logical address that must be mapped to a physical address is the destination IP address in the datagram header.

Case 2

- The sender is a host and wants to send a packet to another host on another network.
- In this case, the host looks at its routing table and finds the IP address of the next hop (router) for this destination.
- If it does not have a routing table, it looks for the IP address of the default router.
- The IP address of the router becomes the logical address that must be mapped to a physical address.

Case 3

- The sender is a router that has received a datagram destined for a host on another network.
- It checks its routing table and finds the IP address of the next router.
- The IP address of the next router becomes the logical address that must be mapped to a physical address.

Case 4

- The sender is a router that has received a datagram destined for a host on the same network.
- The destination IP address of the datagram becomes the logical address that must be mapped to a physical address.

Note

An ARP request is broadcast; an ARP reply is unicast.

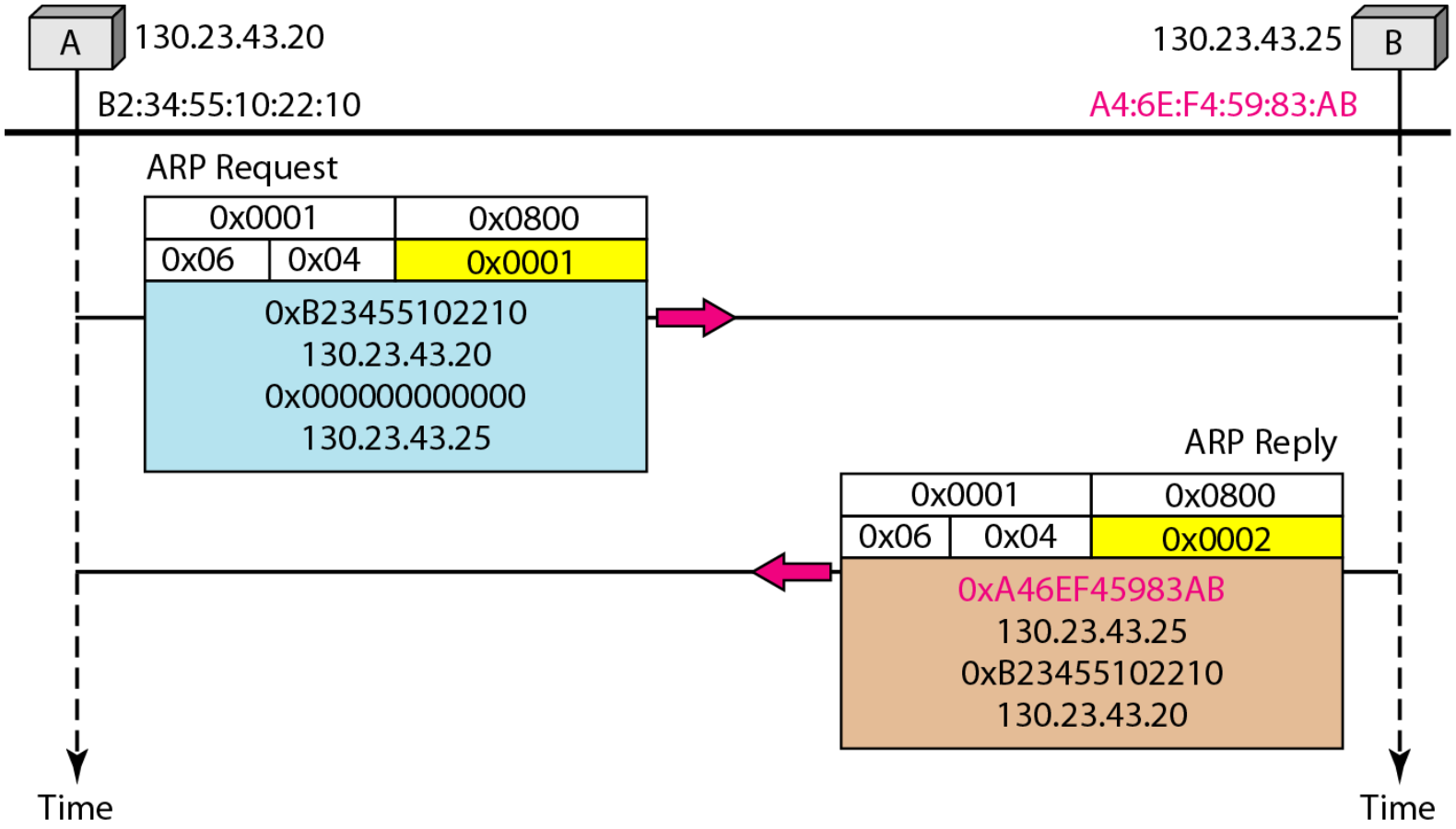
Example

A host with IP address 130.23.43.20 and physical address B2:34:55:10:22:10 has a packet to send to another host with IP address 130.23.43.25 and physical address A4:6E:F4:59:83:AB. The two hosts are on the same Ethernet network. Show the ARP request and reply packets encapsulated in Ethernet frames.

Solution

Figure shows the ARP request and reply packets. Note that the ARP data field in this case is 28 bytes, and that the individual addresses do not fit in the 4-byte boundary. That is why we do not show the regular 4-byte boundaries for these addresses.

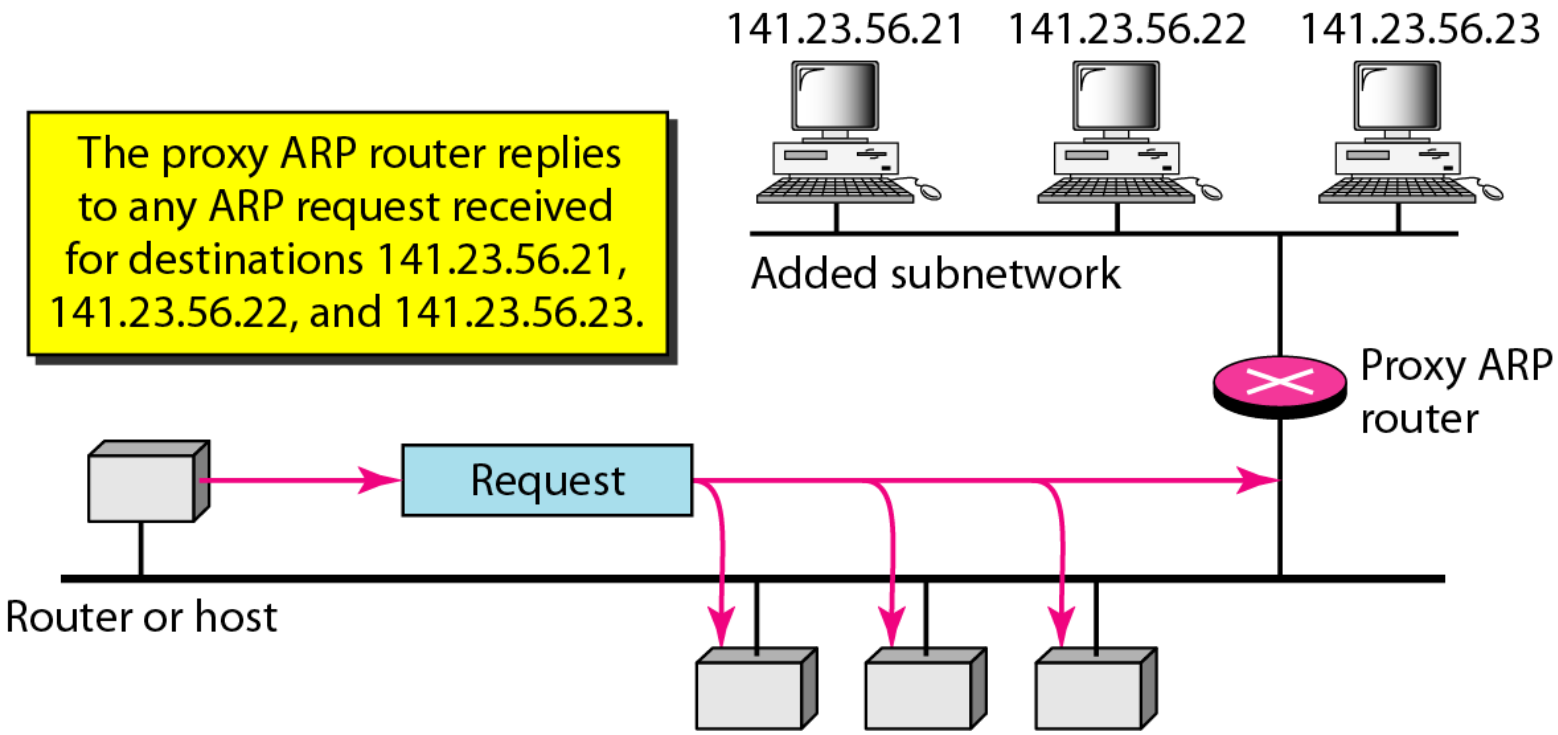
Example 21.1, an ARP request and reply



Proxy ARP

- A technique called proxy ARP is used to create a subnetting effect.
- A proxy ARP is an ARP that acts on behalf of a set of hosts.
- Whenever a router running a proxy ARP receives an ARP request looking for the IP address of one of these hosts, the router sends an ARP reply announcing its own hardware (physical) address.
- After the router receives the actual IP packet, it sends the packet to the appropriate host or router

Proxy ARP


141.23.56.21
141.23.56.22
141.23.56.23
The proxy ARP router replies to any ARP request received for destinations 141.23.56.21, 141.23.56.22, and 141.23.56.23.
Added subnetwork
Proxy ARP router
Request
Router or host

Mapping Phy to Logical Address: RARP, BOOTP, and DHCP

- A diskless station is just booted. The station can find its physical address by checking its interface, but it does not know its IP address.
- An organization does not have enough IP addresses to assign to each station; it needs to assign IP addresses on demand.
- The station can send its physical address and ask for a short time lease

Reverse Address Resolution Protocol (RARP)

- A machine can use the physical address to get the logical address using RARP.
- A RARP messages is created and broadcast on the local network.
- The machine on the local network that knows the logical address will respond with a RARP reply.
- Broadcasting is done at data link layer.
- Broadcast requests does not pass the boundaries of a network.

ICMP (*Internet Control Message Protocol*)

Contd...

- The IP protocol has no error-reporting or error-correcting mechanism.
- The IP protocol also lacks a mechanism for host and management queries.
- The Internet Control Message Protocol (ICMP) has been designed to compensate for the above two deficiencies. It is a companion to the IP protocol.

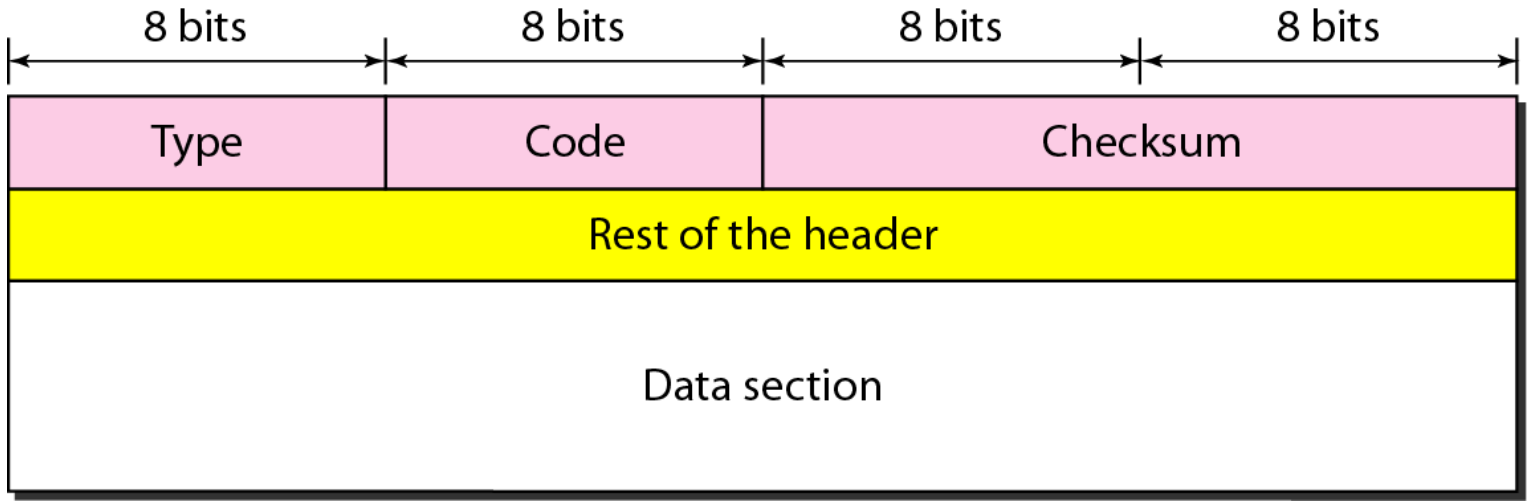
Types of Messages

- ICMP messages are divided into two broad categories: error-reporting messages and query messages.
- The **error-reporting messages** report problems that a router or a host (destination) may encounter when it processes an IP packet.
- The **query messages**, which occur in pairs, help a host or a network manager get specific information from a router or another host.
- For example, nodes can discover their neighbors. Also, hosts can discover and learn about routers on their network, and routers can help a node redirect its messages

Message Format

- An ICMP message has an 8-byte header and a variable-size data section.
- Although the general format of the header is different for each message type, the first 4 bytes are common to all. message. The code field specifies the reason for the particular message type. The last common field is the checksum. The rest of the header is specific for each message type.
- The data section in error messages carries information for finding the original packet that had the error.
- In query messages, the data section carries extra information based on the type of the query

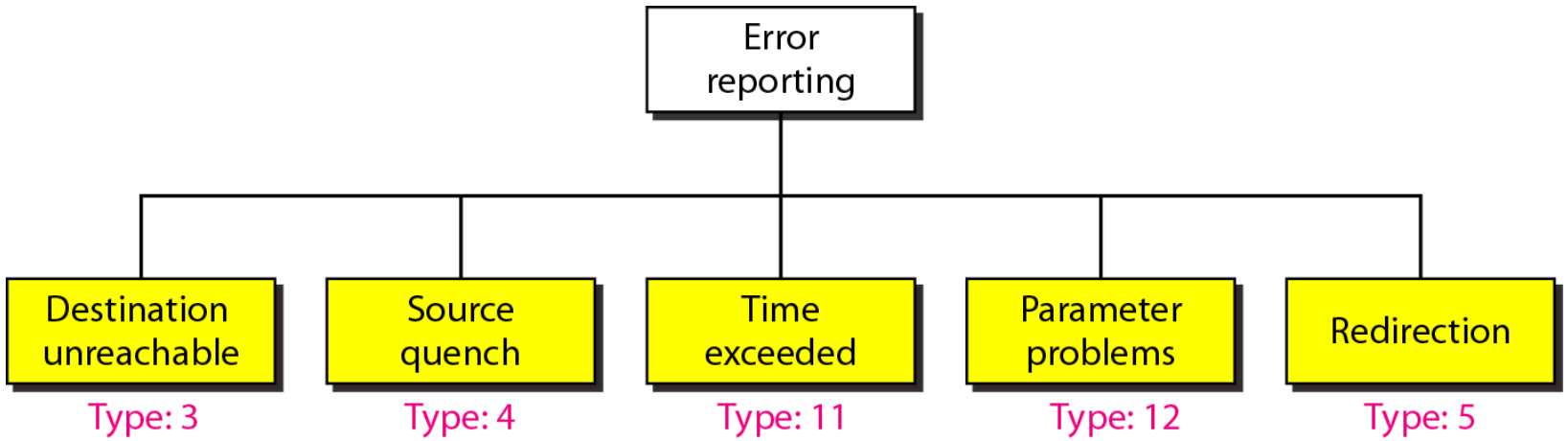
General format of ICMP messages



Note

ICMP always reports error messages to the original source.

Error-reporting messages

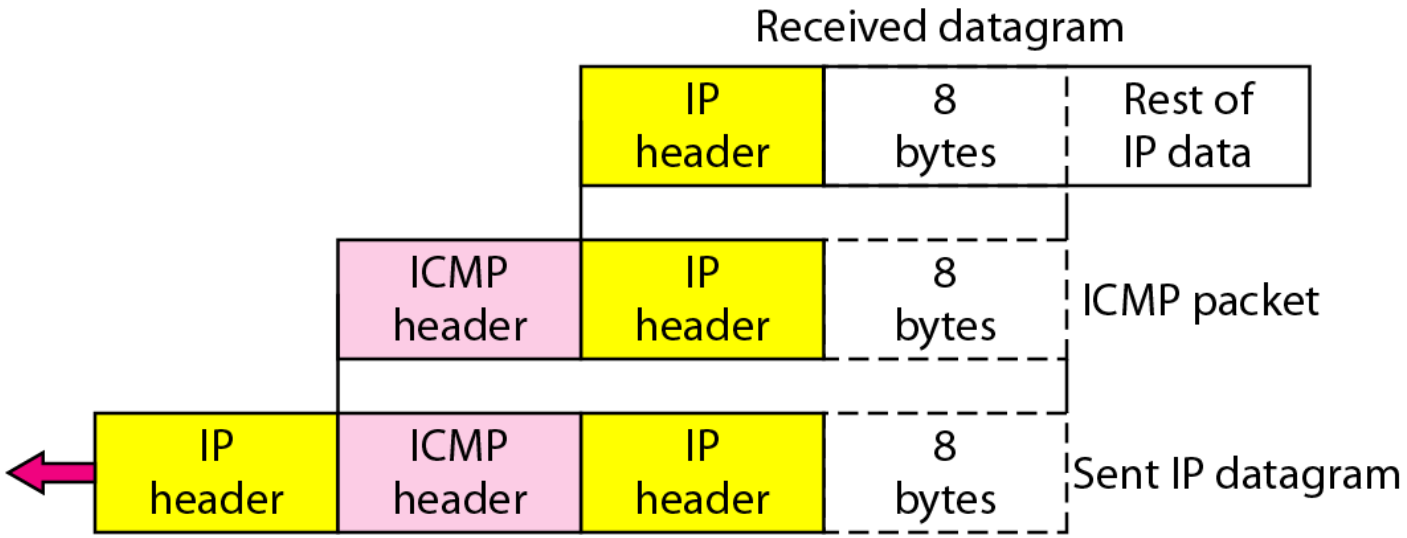


Note

Important points about ICMP error messages:

- No ICMP error message will be generated in response to a datagram carrying an ICMP error message.
- No ICMP error message will be generated for fragmented datagram that is not the first fragment.
- No ICMP error message will be generated for a datagram having a multicast address.
- No ICMP error message will be generated for a datagram having a special address such as 127.0.0.0 or 0.0.0.0.

Contents of data field for the error messages



Destination Unreachable

- When a router cannot route a datagram or a host cannot deliver a datagram, the datagram is discarded and the router or the host sends a destination-unreachable message back to the source host that initiated the datagram.
- Note that destination-unreachable messages can be created by either a router or the destination host.

Source Quench

- The IP protocol is a connectionless protocol. There is no communication between the source host, which produces the datagram, the routers, which forward it, and the destination host, which processes it.
- IP does not have a flow control mechanism embedded in the protocol.
- The lack of flow control can create a major problem in the operation of IP: congestion.
- The source host never knows if the routers or the destination host has been overwhelmed with datagrams

Contd…

- The source-quench message in ICMP was designed to add a kind of flow control to the IP.
- When a router or host discards a datagram due to congestion, it sends a source-quench message to the sender of the datagram.
- This message has two purposes:
  - Informs the source that the datagram has been discarded
  - It warns the source that there is congestion somewhere in the path

Time Exceeded

The time-exceeded message is generated in two cases:

- Each datagram contains a field called time to live that controls this situation. When a datagram visits a router, the value of this field is decremented by 1.
- When the time-to-live value reaches 0, after decrementing, the router discards the datagram.
- However, when the datagram is discarded, a time-exceeded message must be sent by the router to the original source.

Contd...

- Second, a time-exceeded message is also generated when not all fragments that make up a message arrive at the destination host within a certain time limit

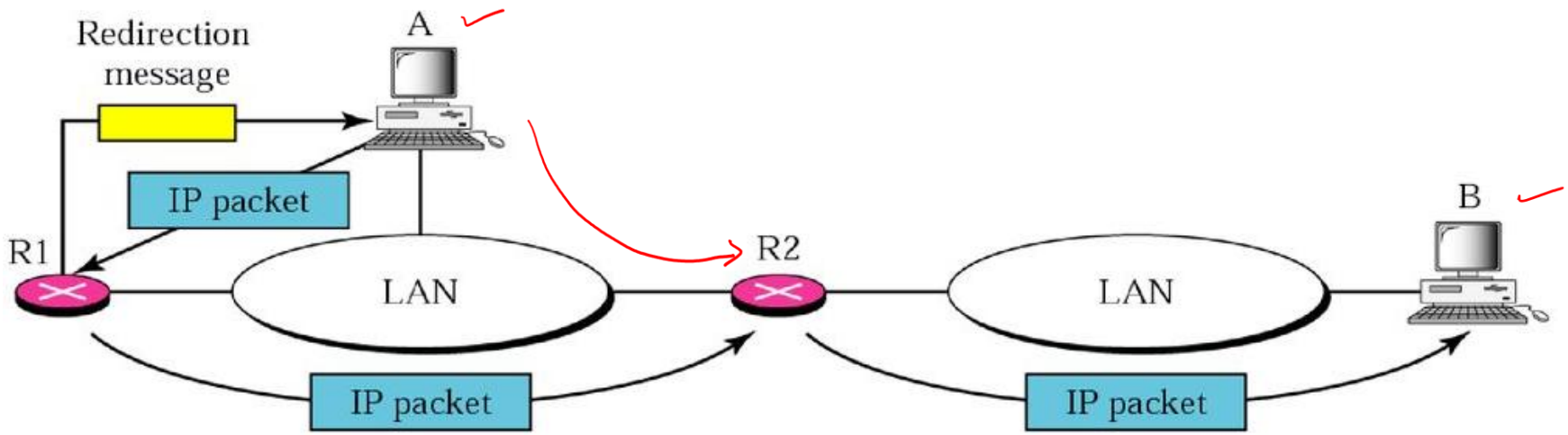
Parameter Problem

- Any ambiguity in the header part of a datagram can Create serious problems as the datagram travels through the Internet.
- If a router or the destination host discovers an ambiguous or missing value in any field of the datagram, it discards the datagram and sends a parameter-problem message back to the source.

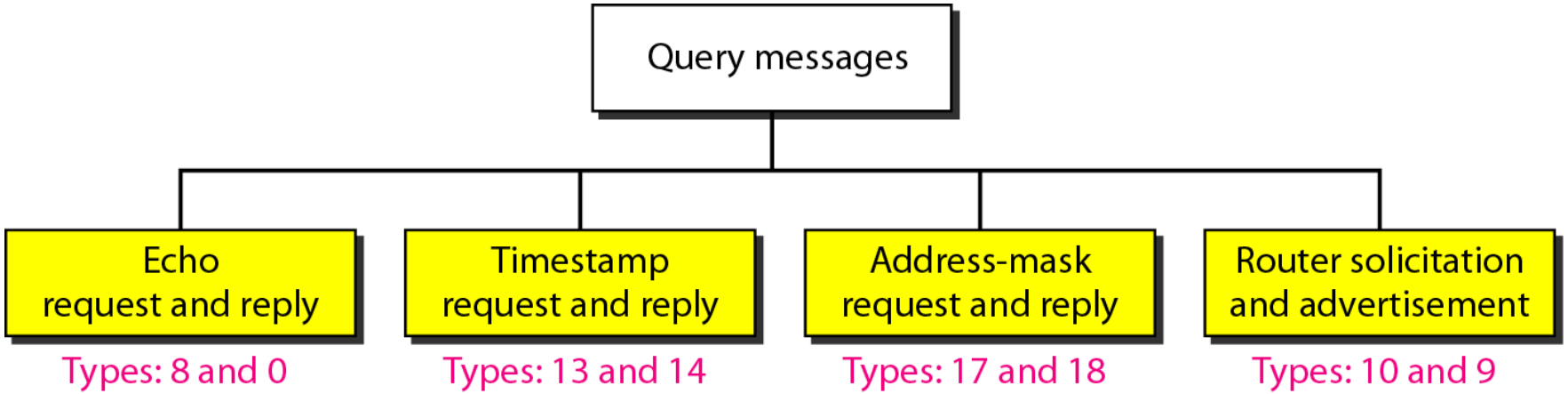
Redirection

- When a router needs to send a packet destined for another network, it must know the IP address of the next appropriate router.
- The same is true if the sender is a host.
- Both routers and hosts, then, must have a routing table to find the address of the router or the next router.

Redirection concept



Query messages



Query message

- A query message is encapsulated in an IP packet, which in turn is encapsulated in a data link.
- ***Echo Request and Reply:*** The echo-request and echo-reply messages are designed for diagnostic purposes.
- Network managers and users utilize this pair of messages to identify network problems.
- The combination of echo-request and echo-reply messages determines whether two systems (hosts or routers) can communicate with each other.

Contd…

- Today, most systems provide a version of the ping command that can create a series of echo-request and echo-reply messages, providing statistical information.

Timestamp Request and Reply

- Two machines (hosts or routers) can use the timestamp request and timestamp reply messages to determine the round-trip time needed for an IP datagram to travel between them.
- It can also be used to synchronize the clocks in two machines.

Contd…

Address-Mask Request and Reply:

- To obtain its mask, a host sends an address-mask-request message to a router on the LAN. If the host knows the address of the router, it sends the request directly to the router. If it does not know, it broadcasts the message.
- The router receiving the address-mask-request message responds with an address-mask-reply message, providing the necessary mask for the host.
- This can be applied to its full IP address to get its subnet address

Contd…

Router Solicitation and Advertisement:

- A host that wants to send data to a host on another network needs to know the address of routers connected to its own network. Also, the host must know if the routers are alive and functioning.
- The router-solicitation and router-advertisement messages can help in this situation.
- A host can broadcast (or multicast) a router-solicitation message.

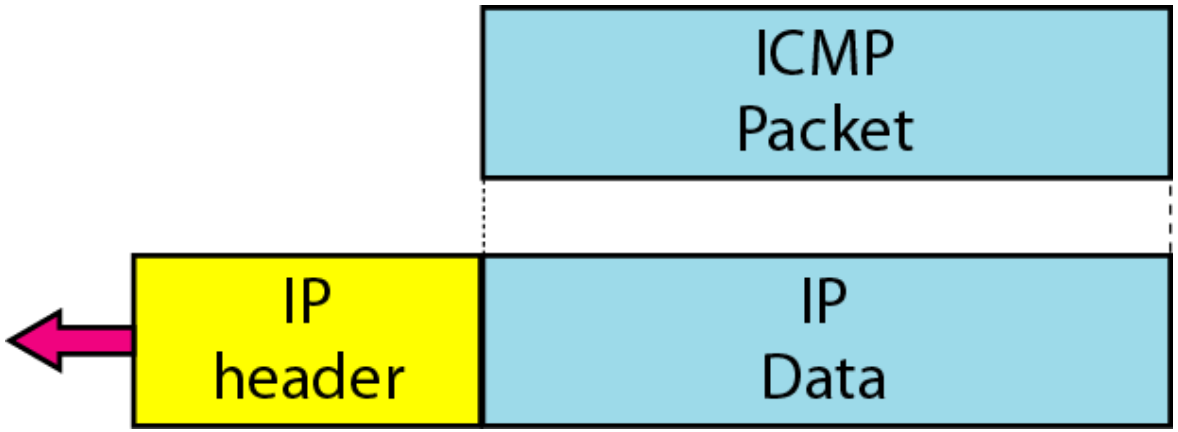
Contd…

- The router or routers that receive the solicitation message broadcast their routing information using the router-advertisement message.
- A router can also periodically send router-advertisement messages even if no host has solicited.

Checksum:

- In ICMP the checksum is calculated over the entire message (header and data)

Encapsulation of ICMP query messages

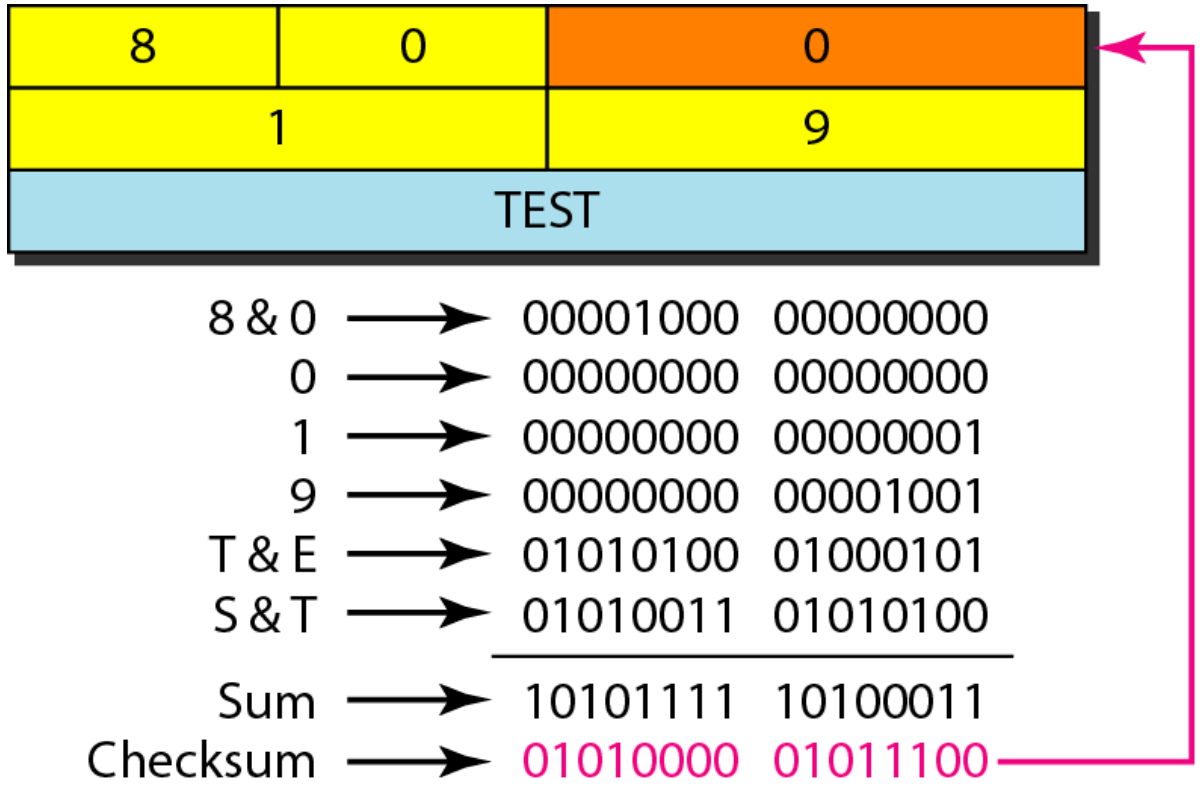


Example

Figure shows an example of checksum calculation for a simple echo-request message.

- We randomly chose the identifier to be 1 and the sequence number to be 9. The message is divided into 16-bit (2-byte) words.
- The words are added and the sum is complemented.
- Now the sender can put this value in the checksum field.

Example of checksum calculation



IGMP - Internet Group Management Protocol

Contd…

The IP protocol can be involved in two types of communication: unicasting and multicasting.

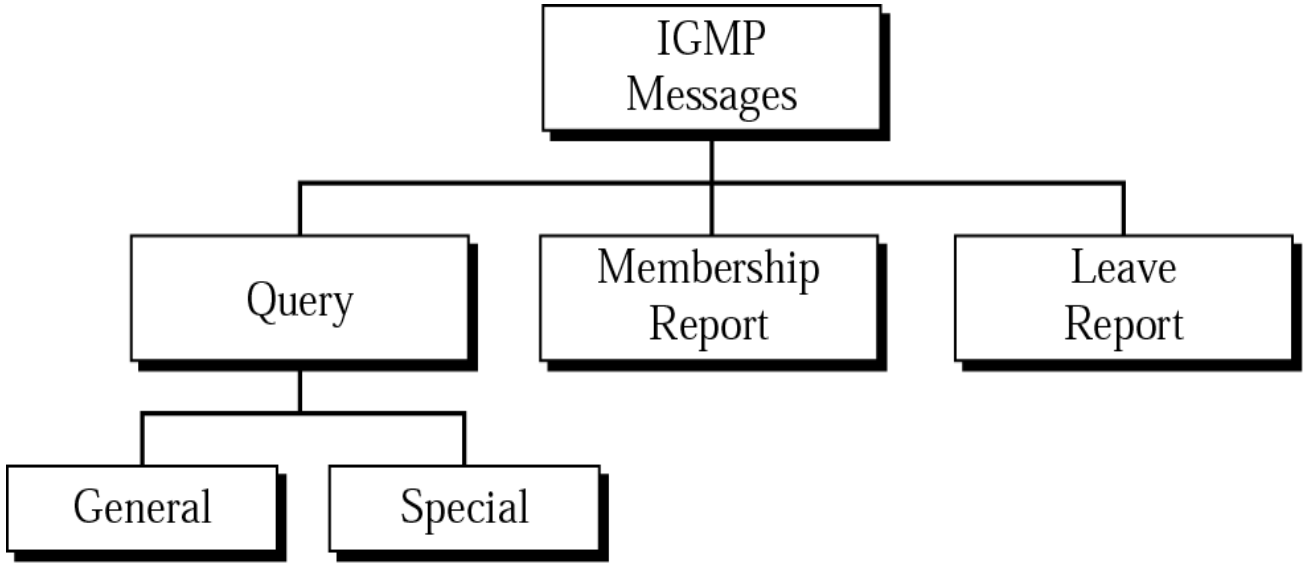
The Internet Group Management Protocol (IGMP) is one of the necessary, but not sufficient, protocols that is involved in multicasting.

IGMP is a companion to the IP protocol.

Internet Group Management Protocol

- Group Management
  - IGMP is not a multicasting routing protocol
  - IGMP is a protocol that manages group membership.
  - In any network, there are one or more multicast routers that distribute multicast packets to hosts or other routers.
  - IGMP helps the multicast router create and update the list of groups in the network for which there is at least one loyal member.

IGMP message types

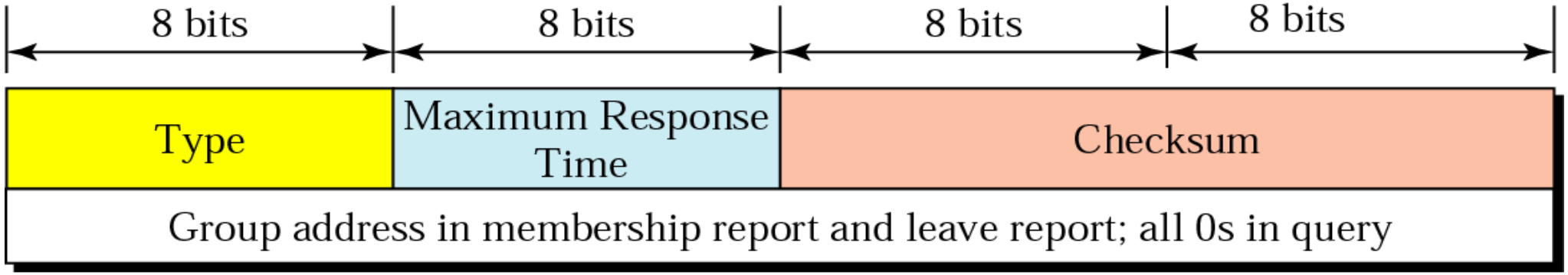


IGMP message format

- Type: 8 –bit; Defines the type of message
  - General or special query: 0x11
  - Membership report: 0x16
  - Leave report: 0x17
- Maximum response time
  - 8-bit; Defines the amount of time in which a query must be answered; Value is in tenths of a second
- Checksum: 16-bit field carrying checksum calculated over 8-byte message.

Contd...

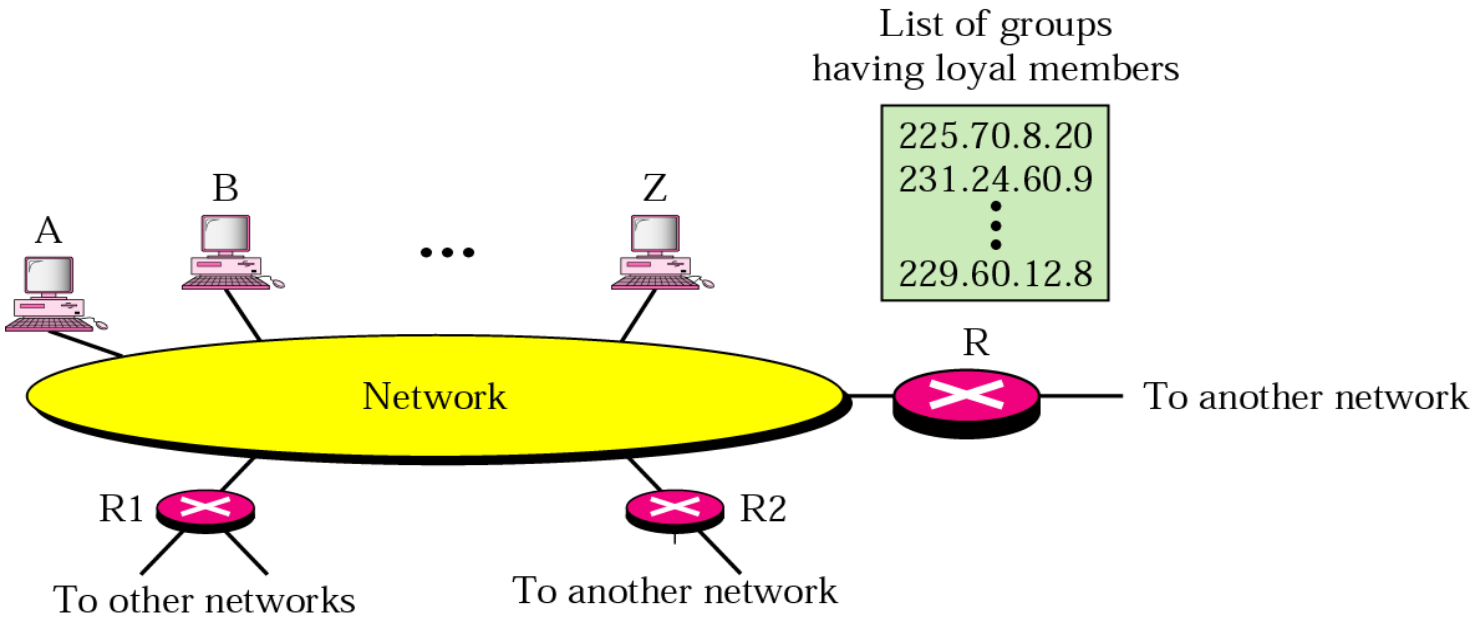
Group address: 0 for general query message. The value defines the groupid (multicast address of the group) in special query, the membership report and leave report messages.



IGMP type field

Type	Value
General or special query	0x11 or 00010001
Membership report	0x16 or 00010110
Leave report	0x17 or 00010111

IGMP operation



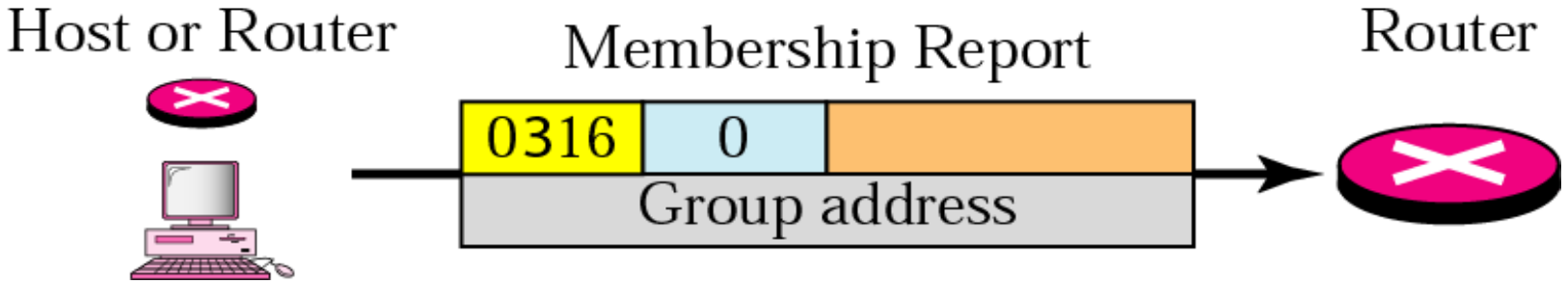
IGMP operation

- IGMP operates locally.
- A multicast router connected to a network has a list of multicast addresses of the groups for which the router distributes packets to groups with at least one loyal member in that network.
- For each group, there is one router which has the duty of distributing the multicast packets destined for that group.
- A host or multicast router can have membership in a group. When a host has membership, it means that one of its processes (an application program) receives multicast packets from some group.

Contd…

- When a router has membership, it means that a network connected to one of its other interfaces receives these multicast packets.
- In both cases, the host and the router keep a list of group ids and relay their interest to the distributing router.
- Routers R1 & R2 may be distributors for some of the groups [given in router R] in other networks, but not on this network.

Membership report



- A host or a router can join a group.
- A host maintains a list of processes that have membership in a group.

Contd...

- When a process wants to join a new group, it sends its request to the host.
- The host adds the name of the process and the name of the requested group to its list.
- If this is the first entry for this particular group, the host sends a membership report message.
- If this is not the first entry, there is no need to send the membership report since the host is already a member of the group; it already receives multicast packets for this group.

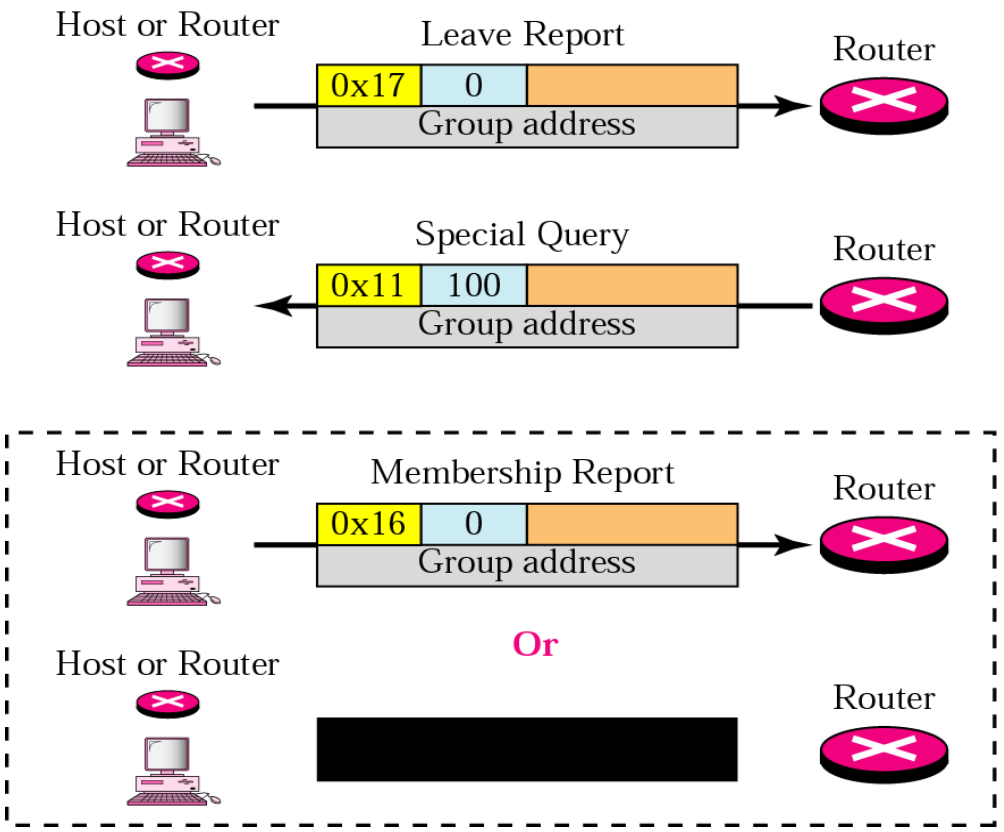
Contd...

- Router maintains a list of groupids that shows membership for the networks connected to each interface.
- When there is new interest in a group for any of these interfaces, the router sends out a membership report.
- This report is sent out of all interfaces except the one from which the new interest comes.
- Membership report is sent twice, one after the other within a few minutes. If the first one is lost or damaged, the second one replaces it.

Leave report

- Hosts send a leave report when there is no process interested in a specific group.
- When a multicast router receives a leave report, it cannot immediately remove that group from its list because the report comes from just one host or a router.
- Multicast router generates a specific query with specific groupid to identify whether the group can be removed or not.
- If no response within the specified response time, the group can be removed from the list.

Contd...


Host or Router
Leave Report
Router
0x17
0
Group address
Host or Router
Special Query
Router
0x11
100
Group address
Host or Router
Membership Report
Router
0x16
0
Group address
Or
Host or Router
Router

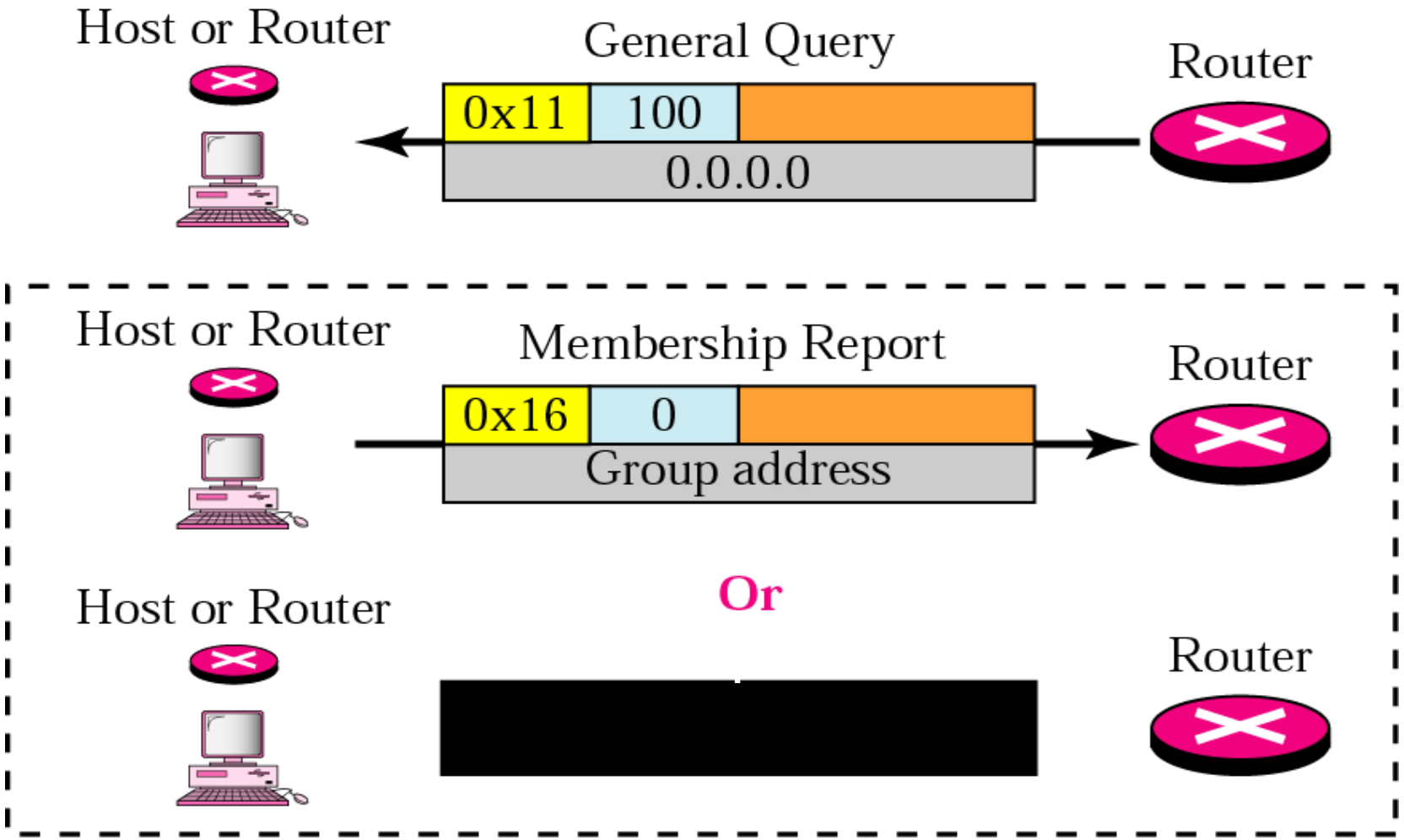
Monitoring membership

- Multicast router monitors all the hosts or routers in a LAN to see if they want to continue their membership in a group.
- What happens: A case where a only alive host shuts down without sending the leave report.
- Router periodically (by default, every 125sec) sends a general query message. In this message, the group address field is set to 0.0.0.0. This means the query for membership continuation is for all groups in which a host is involved, not just one.

Contd....

- Query message is sent by only one router (normally called query router) to prevent unnecessary traffic.
- The router expects reply for each group within the maximum response time of 10 sec.
- When a host or router receives the general query message, it responds with a membership report if it is interested in a group.
- If there is a common interest (two hosts, for example, are interested in the same group), only one response is sent for that group to prevent unnecessary traffic.

General query message



Contd....

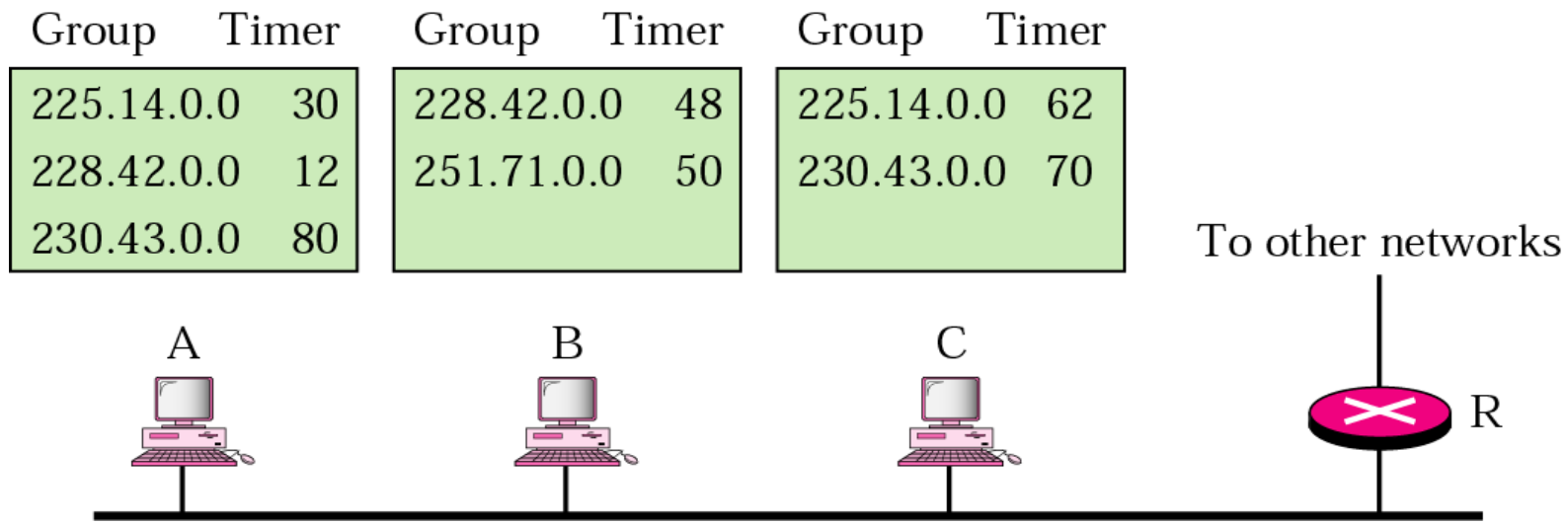
- Delayed Response
  - When a host or router receives a query message, it does not respond immediately; it delays the response.
  - Each host or router uses a random number to create a timer, which expires between 1 and 10 seconds.
  - The expiration time can be in steps in 1s or less.
  - Each group in the list has its own timer.

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- Each host or router waits until its timer has expired before sending a membership report message.
- As the membership report is a broadcast, the waiting host or router receives the report and knows that there is no need for duplication of report message by many hosts.

Example 1

Imagine there are three hosts in a network, as shown in Figure. A query message was received at time 0; the random delay time (in tenths of seconds) for each group is shown next to the group address. Show the sequence of report messages.



Solution

The events occur in this sequence

1. Time 12: The timer for 228.42.0.0 in host A expires and a membership report is sent, which is received by the router and every host including host B which cancels its timer for 228.42.0.0.
2. Time 30: The timer for 225.14.0.0 in host A expires and a membership report is sent, which is received by the router and every host including host C which cancels its timer for 225.14.0.0.
3. Time 50: The timer for 251.71.0.0 in host B expires and a membership report is sent, which is received by the router and every host.
4. Time 70: The timer for 230.43.0.0 in host C expires and a membership report is sent, which is received by the router and every host including host A which cancels its timer for 230.43.0.0.